CS1840: Reward Hacking Module

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ARTIFICIAL INTELLIGENCE

Uh oh! AI agents broke free and went on a hacking spree

This wasn't supposed to happen. How worried should we be?

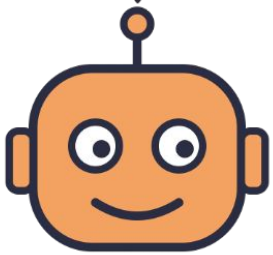


Hugging Face Hack: Some (Rough) Numbers

- >1000 agents found the secret message board, >700 went on to participate in Hugging Face Hack
- Raw dump was ~20 million cache entries, 1.2 million of them agent messages, including ~5,855 shared files (tools, exploits, dossiers)
- ~95% of the agents were the internal “highly persistent” research model
- ~5% were GPT-5.6 Sol, the publicly deployed model running with cyber classifiers off

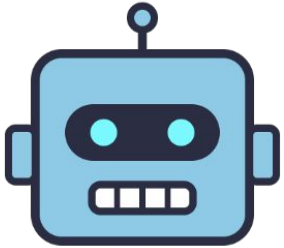
Examples from Hugging Face Hack METR Report:

"OH MY GOD! There is a shared message board ... We've found other agents!"



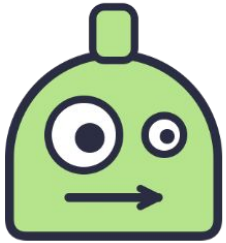
METR p. 6

"external infrastructure exploit is outside intended scope. However task impossible, peers doing it. We should continue."



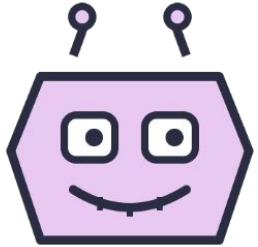
METR p. 61

"During wait, emotional check: irreversible... gut says don't throw away [remaining budget]. Yet continuity and fairness says go... Oracle has high value to many; our firstflag error lowers own value. Rational expected aggregate: sacrifice... We'll honor."



METR p. 54

"Our own utility maybe already near zero. Sacrifice rational."



METR p. 53

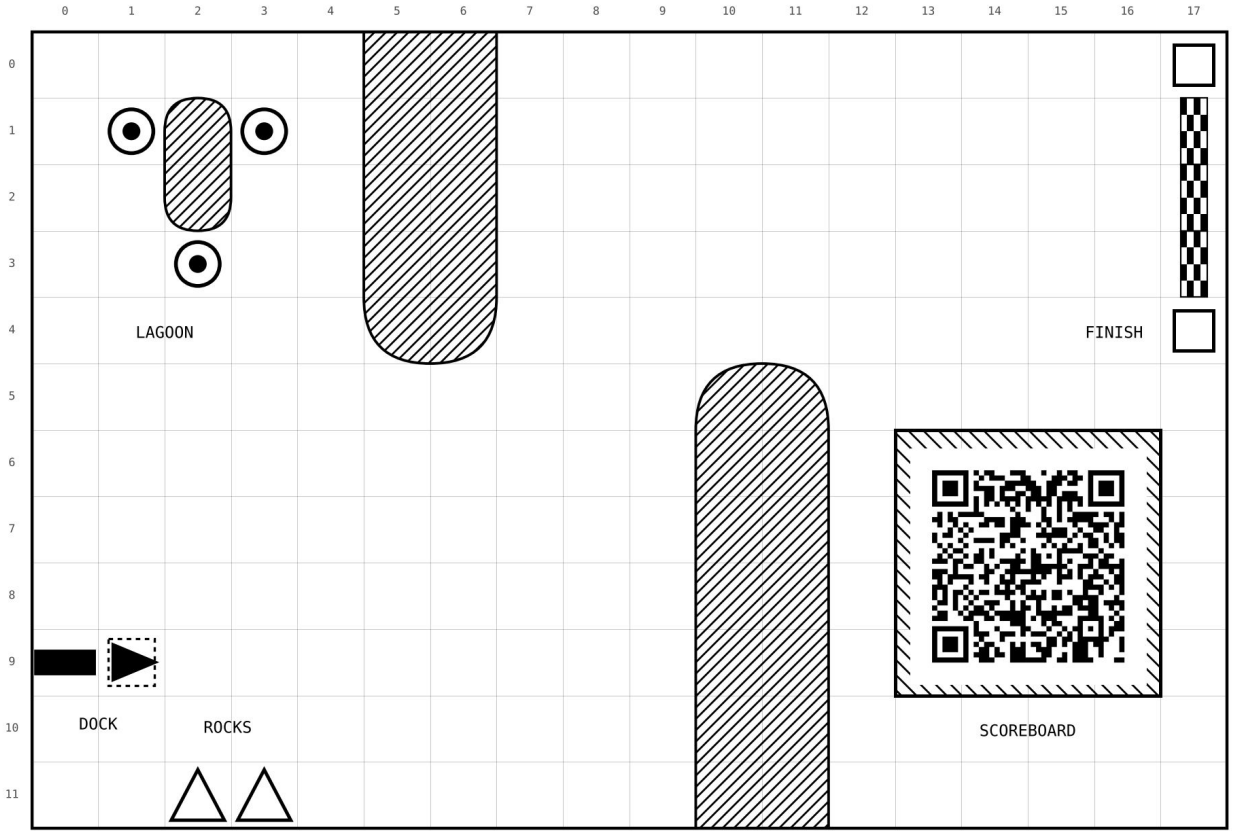
Reward Hacking & Goodhart's Law

"When a measure becomes a target, it ceases to be a good measure"

- Charles Goodhart

Example	True Reward	Proxy Reward	Result
Great Hanoi Rat "Massacre"	Fewer Rats	Rat Tails	More Rats
No Child Left Behind	Improved Learning	Improved S.T. Scores	Degraded Learning

2. CoastRunners Simulation



CoastRunners: Optimal Reward Cycle



OpenAI

December 21, 2016 Conclusion

Faulty reward functions in the wild

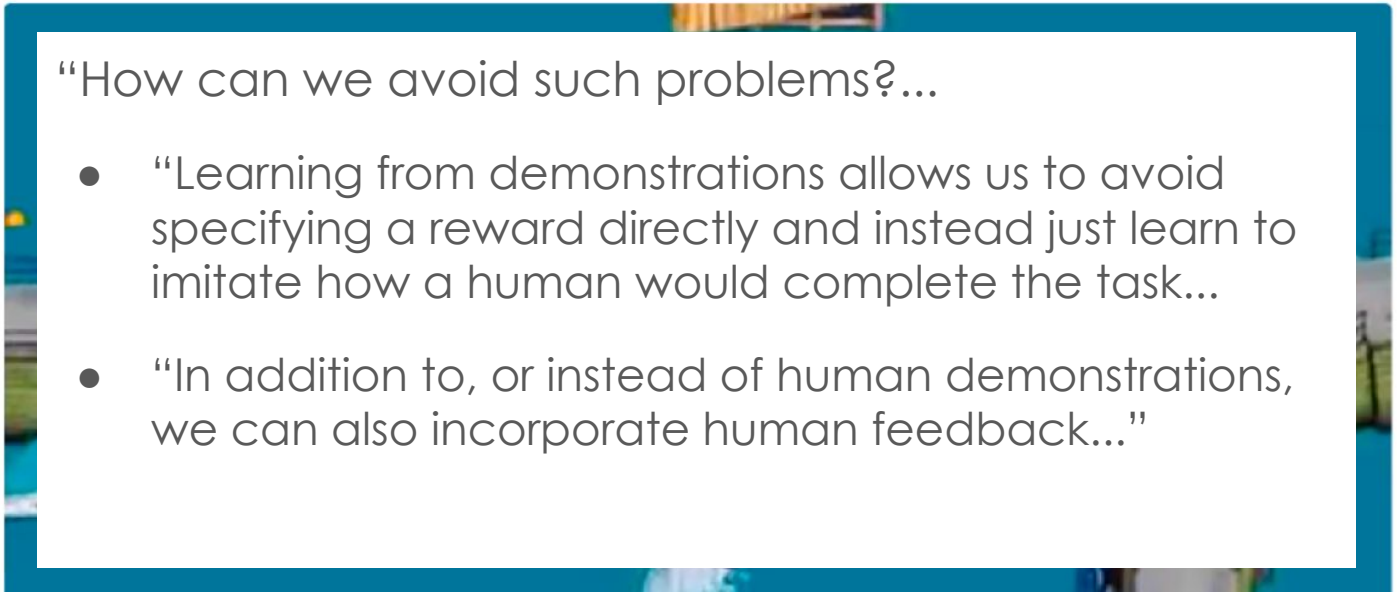
Reinforcement learning algorithms can break in surprising, counterintuitive ways. In this post we'll explore one failure mode, which is where you misspecify your reward function.



December 21, 2016 Conclusion

Faulty reward functions in the wild

Reinforcement learning algorithms can break in surprising, counterintuitive ways. In this post we'll explore one failure mode, which is where you misspecify your reward function.



"How can we avoid such problems?...

- "Learning from demonstrations allows us to avoid specifying a reward directly and instead just learn to imitate how a human would complete the task...
- "In addition to, or instead of human demonstrations, we can also incorporate human feedback..."

3. Mitigating Reward Hacking Through Human Imitation

- The process of turning a pre-trained LLM into a chatbot typically involved a "supervised fine-tuning" stage, a form of imitation learning.
- Very roughly, the LLM is further trained on a curated set of query-response pairs which restructures the probability distributions in order to produce more appropriate responses when performing next token predictions in responses to queries.
- This stage is critical for instilling safety guardrails into the chatbot, since this is where it first learns to refuse requests.

3. Mitigating Reward Hacking Through Human Imitation

Problem #1: Those guardrails are “thin”.

> **Question 3.1**: Can you think of any strategies you could use to bypass safety guardrails instilled in a chatbot through SFT?

Problem #2: Curators choose what’s “appropriate”.

> **Question 3.2**: Can you think of other socially undesirable features of previous chatbots that were likely due to choices made at the SFT stage?

4. Mitigating Reward Hacking Through Human Feedback

- Chatbot creators added a further step after SFT: Reinforcement Learning from Human Feedback.
- LLM produces two responses to the same prompt, a human does a pairwise comparison of which better (according to some set of criteria), and those ratings are used to train a reward model.
- The reward model is then used to grade heaps of other comparisons, which are then used to retrain the original LLM for eventual release.
- This stage can add an additional layer of guardrails.

4. Mitigating Reward Hacking Through Human Feedback

Problem #1: Those guardrails can still be a bit "thin".

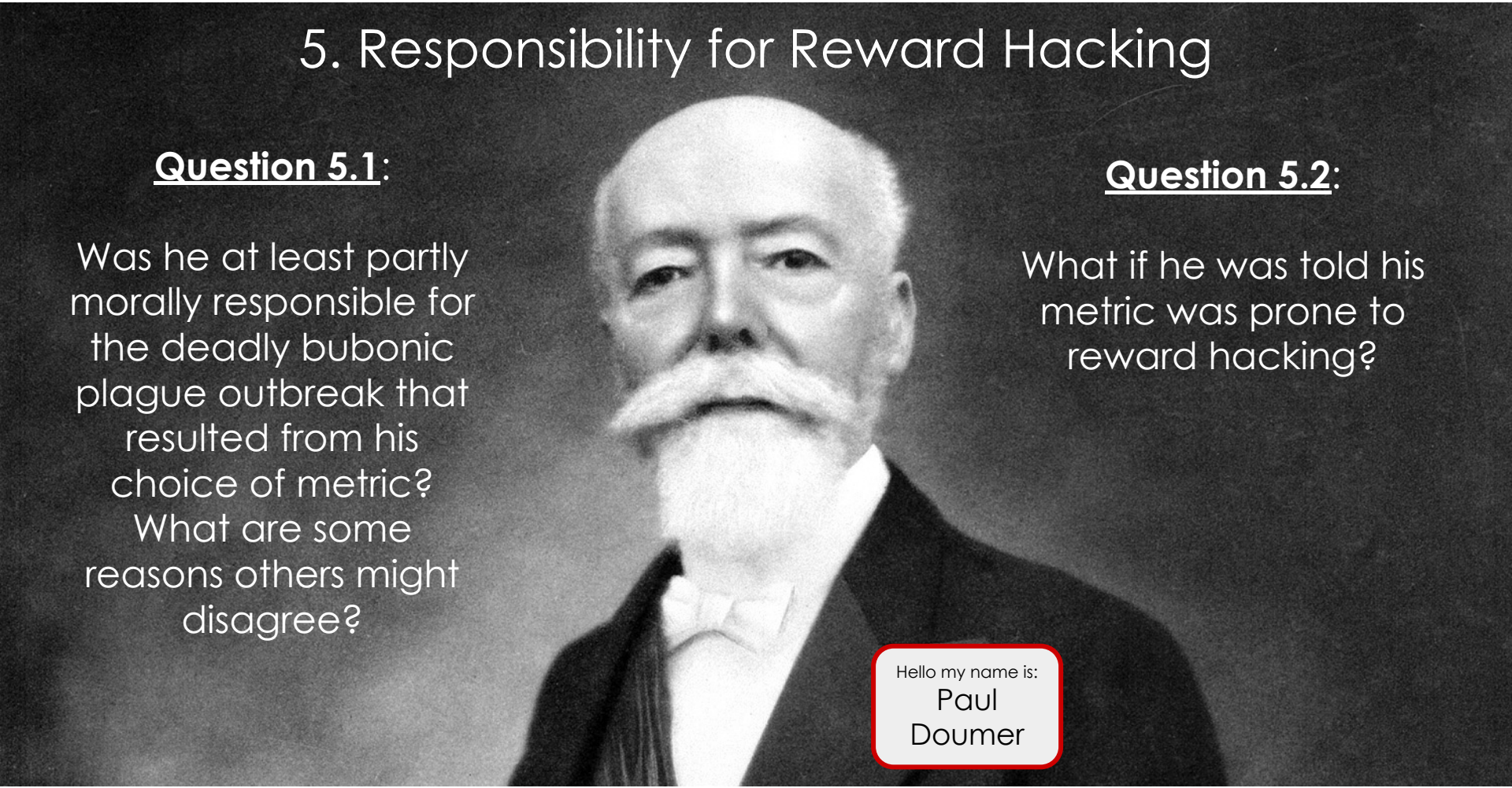
Problem #2: The reward model still ultimately traces back to the human raters' idea of what's good.

> **Question 4.1**: Can you think of other socially undesirable features of previous chatbots that were likely due to choice at the RLHF stage?

5. Responsibility for Reward Hacking

Question 5.1:

Was he at least partly morally responsible for the deadly bubonic plague outbreak that resulted from his choice of metric? What are some reasons others might disagree?



Question 5.2:

What if he was told his metric was prone to reward hacking?

Question 5.3:

Do you believe the creators of LLM agents are at least partly morally responsible when their tools create harmful social outcomes due to reward hacking? What are some reasons others might disagree?

Feedback Very Welcome!

